

# Luca Lazzaroni

**Ricercatore a tempo determinato in Elettronica**

DITEN – UNIGE

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SETTORE RTD-A · IINF-01/A

BIBLIOMETRIA — SCOPUS

| 68 | 345 | 11 | 21 |
|---|---|---|---|
| DOCUMENTI | CITAZIONI | INDICE H | ARTICOLI, 5 ANNI |

AGG. 26 OTT 2026

## 01 Profilo

Ricercatore a tempo determinato (RTD-A) presso il Dipartimento di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni (DITEN), Università di Genova, settore IINF-01/A – Elettronica, dal marzo 2025. Dottore di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni dal 2024.

Attività di ricerca su machine learning embedded ed edge AI: toolchain TinyML, reti neurali quantizzate e binarie su microcontrollori e FPGA, percezione e decisione per la guida automatizzata, deep reinforcement learning, LLM on-device. L'attività si svolge anche nell'ambito di progetti di ricerca europei e nazionali con partner industriali.

Edge AI · TinyML · Deep learning embedded · Guida automatizzata · Deep reinforcement learning · LLM on-device

## 02 Contatti e profili

| | | |
|---|---|---|
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| SCOPUS 57220892898 | GOOGLE SCHOLAR w_6a6sYAAAAJ | GITHUB @lucalazzaroni |
| LINKEDIN luca-lazzaroni | | |

## 03 Incarichi e formazione

**POSIZIONI**

Mar 2025 – oggi

**Ricercatore a tempo determinato (RTD-A)**

DITEN, Università di Genova

Settore scientifico-disciplinare IINF-01/A — Elettronica.

Ott 2023 — Feb 2025

**Assegnista di ricerca**

DITEN, Università di Genova

Assegno: “Realizzazione di modelli di Machine Learning per il trattamento di dati provenienti da veicoli autonomi”.

Nov 2020 — Ott 2023

**Dottorando di ricerca**

DITEN, Università di Genova — XXXVI ciclo

Borsa di dottorato: “Deep learning on the edge”.

### FORMAZIONE

2024

**Dottorato di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni**

Università di Genova — conseguito il 14 marzo 2024

Tesi: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”

2020

**Laurea Magistrale in Ingegneria Elettronica (LM-29)**

Università di Genova — 24 luglio 2020

Tesi: “Progetto e implementazione di un motore di esecuzione multipiattaforma per applicazioni IoT” · Voto: 110/110

2017

**Laurea in Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)**

Università di Genova — 27 ottobre 2017

Tesi: “Progetto model-based del controllo di robot Lego in ambiente Simulink”

2014

**Diploma di Maturità Scientifica**

Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV)

Voto: 100/100

Abilitazione all'esercizio della professione di Ingegnere dell'Informazione — Esame di Stato superato nella seconda sessione 2020.

## 04 Didattica e incarichi accademici

### CORSI

A.A. 2024/25 – 2026/27

**Distributed Electronic Systems and Technologies for Environmental Monitoring**

Docente titolare · cod. 94631

A.A. 2024/25 – 2026/27

**Tiny Machine Learning**

Co-titolare · cod. 114752

A.A. 2026/27

**Dispositivi e Circuiti Elettronici**

Docente titolare · cod. 65997

dall'A.A. 2022/23

**Machine Learning for Automated Driving**

Cultore della materia · cod. 108716

| | |
|---|---|
| A.A. 2023/24 | **Computer Graphics Arch. — Games and Simulation**<br>Docente a contratto · cod. 172317 |
| A.A. 2022/23 | **Autonomous Agents in Games**<br>Supporto alla didattica · cod. 198216 |
| A.A. 2020/21 | **Fondamenti di Programmazione di Sistemi Elettronici**<br>Supporto alla didattica · cod. 166061 |
| A.A. 2020/21 | **Approccio Makers alla Progettazione Elettronica**<br>Supporto alla didattica · cod. 186985 |

Tutti i corsi presso l'Università di Genova.

### SUPERVISIONE

**1** Dottorando — Ali Zare, Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni (XLI ciclo)

**12** Tesi di Laurea — Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)

**10** Tesi di Laurea Magistrale — Ingegneria Elettronica (LM-29)

**7** Tesi di Laurea Magistrale — Ingegneria Tecnologica per la Strategia e la Sicurezza (LM/DS)

**2** Tesi di Laurea Magistrale — Informatica (LM-18)

### INCARICHI ACCADEMICI

**Membro della Commissione Didattica e Ammissione del Corso di Laurea Magistrale in Engineering for Natural Risk Management.**

**Responsabile della comunicazione per il Corso di Laurea Magistrale in Electronic Engineering.**

## 05 Curatela e servizio

### RUOLI EDITORIALI

2026 — **Editor del volume**
APPLEPIES 2026 — Springer LNEE
Atti in corso di pubblicazione.

2025 — **Membro dell'Editorial Board (Early Career)**
Sensors — MDPI
Dal 18 dicembre 2025.

2025 — **Editor del volume**
APPLEPIES 2025 — Springer LNEE
Atti della International Conference on Applications in Electronics Pervading Industry, Environment and Society.

2024

**Guest Editor**

Energies — MDPI

Special Issue “Tiny Machine Learning for Energy Applications”, aprile 2024 — agosto 2025.

## ORGANIZZAZIONE DI CONVEGNI

**Program Chair**

APPLEPIES 2026

Bologna, Italia, 10–11 settembre 2026.

**Program Chair**

APPLEPIES 2025

Torino, Italia, 11–12 settembre 2025.

**Program Committee**

APPLEPIES

Dal 2023.

**Program Committee**

GALA — Games and Learning Alliance

Dal 2023.

**Chairman di sessione**

APPLEPIES 2023

1st Short Oral session — Genova, Italia, 28–29 settembre 2023.

## ATTIVITÀ DI REVISIONE

### RIVISTE

| | |
|---|---|
| IEEE Access | dal 2023 |
| Journal of Systems Architecture | dal 2023 |
| Engineering Applications of Artificial Intelligence | dal 2024 |
| Sensors | dal 2024 |
| IET Intelligent Transport Systems | dal 2025 |
| IEEE Transactions on Intelligent Vehicles | dal 2026 |
| IEEE Transactions on Circuits and Systems II: Express Briefs | dal 2026 |
| Robotics and Autonomous Systems | dal 2026 |
| Scientific Reports | dal 2026 |
| Journal of Big Data | dal 2026 |

### CONFERENZE

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|---|---|
| IEEE ICRA — International Conference on Robotics and Automation | dal 2026 |
| IEEE/RSJ IROS — Intelligent Robots and Systems | dal 2025 |
| GALA — Games and Learning Alliance | dal 2023 |
| APPLEPIES | dal 2021 |

Discover Internet of Things dal 2026

## 06 Interventi a convegni

13 INTERVENTI

| | |
|---|---|
| 11–12 set 2025 | **APPLEPIES 2025**<br>Torino, Italia |
| 11–14 gen 2025 | **43rd IEEE ICCE — International Conference on Consumer Electronics**<br>Las Vegas, NV, USA |
| 19–20 set 2024 | **APPLEPIES 2024**<br>Torino, Italia |
| 24–26 giu 2024 | **55th Annual Meeting of the Italian Society of Electronics (SIE)**<br>Genova, Italia |
| 28–29 set 2023 | **APPLEPIES 2023**<br>Genova, Italia |
| 7–9 set 2023 | **54th Annual Meeting of the Italian Society of Electronics (SIE)**<br>Noto (RG), Italia |
| 26–27 set 2022 | **APPLEPIES 2022**<br>Genova, Italia |
| 7–9 set 2022 | **53rd Annual Meeting of the Italian Society of Electronics (SIE)**<br>Pizzo (VV), Italia |
| 21–22 set 2021 | **APPLEPIES 2021**<br>Pisa, Italia |
| 1–2 dic 2021 | **GALA 2021 — Games and Learning Alliance**<br>La Spezia, Italia (online) |
| 7–9 lug 2021 | **52nd Annual Meeting of the Italian Society of Electronics (SIE)**<br>Trieste, Italia (online) |
| 9–10 dic 2020 | **GALA 2020 — Games and Learning Alliance**<br>Laval, Francia (online) |
| 19–20 nov 2020 | **APPLEPIES 2020**<br>Online |

## 07 Premi e associazioni

2025

**Best Presentation of the Session — sessione “Edge Computing”**

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 gennaio 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

### ASSOCIAZIONI PROFESSIONALI

**IEEE**

IEEE Member e IEEE Young Professional.

**IEEE Consumer Technology Society**

Membro del Technical Committee “Machine Learning, Deep Learning and AI in CE (MDA)”.

**SIE — Società Italiana di Elettronica**

Socio nell'Unità di Ricerca dell'Università di Genova dal 2021, area di ricerca “Sistemi elettronici ed applicazioni”.

## 08 Progetti di ricerca

Synergies

**Generation of scenarios for development, training, virtual testing and validation of CCAM systems**

Europeo — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework: T8.1 e T8.4 (Future Planning)

Genera scenari di test per addestramento, test e validazione di sistemi CCAM, con focus su traffico urbano ed extraurbano — dove la conoscenza degli scenari rilevanti è molto più scarsa che in autostrada. Coinvolge le principali case automobilistiche europee e molte università del settore.

Hi-Drive

**Addressing challenges toward the deployment of higher automation**

Europeo — H2020, GA 101006664 · Membro dell'unità di ricerca dell'Università di Genova

Sviluppa e testa su larga scala funzionalità di guida automatizzata avanzata, passando dall'automazione condizionata (SAE L3) a quella alta, in diversi scenari e contesti stradali.

L3Pilot

**Piloting Automated Driving on European Roads**

Europeo — H2020 · Membro dell'unità di ricerca dell'Università di Genova

Sperimentazione su larga scala di funzioni di guida automatizzata SAE L3 su strade pubbliche europee, con le principali case automobilistiche e fornitori europei.

GEMINI

**Green Machine Learning for the IoT**

Nazionale — PRIN 2022 · Membro dell'unità di ricerca dell'Università di Genova (DII–UNITN / DITEN–UNIGE)

Un framework per applicazioni TinyML pienamente sostenibili distribuite lungo un continuum edge-to-cloud, pensato per dispositivi embedded senza batteria, inclusi quelli privi di connettività di rete.

ADINA

**Assistente Digitale Integrato Automotive**

Regionale — POR FESR Piemonte 2014-2020 · Membro dell'unità di ricerca dell'Università di Genova, con Lifetouch

Un assistente virtuale integrato su hardware dedicato a bordo vettura e connesso ai sistemi del veicolo: algoritmi di IA interpretano le richieste del guidatore e agiscono su funzioni veicolo e infotainment.

## 09 Pubblicazioni

74 VOCI · 68 SU SCOPUS

2026

**Cross-platform deployment and characterization of an on-device generative conversational agent**
P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonesi, D.P. Pau
*Journal of Systems Architecture* vol. 181, art. 103981 Rivista

**A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction**
L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti
*IEEE Signal Processing Letters* vol. 33, pp. 136–140 Rivista

**Efficient Certified Robustness Assessment for Transformer Models via Verifier-Aware Perturbation-Position Scheduling**
V. Soltanmuradov, R. Berta, **L. Lazzaroni**, D.M. Gómez, A. Pighetti, F. Bellotti
*Machine Learning and Knowledge Extraction* vol. 8, no. 9, art. 290 Rivista

**FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction**
A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti
*IEEE Transactions on Circuits and Systems II: Express Briefs* vol. 73, no. 8, pp. 888–892 Rivista

**Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms**
H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta
*IEEE Access* vol. 14, pp. 44886–44903 Rivista

**StabilEdge, Formal Verification of Edge AI Applications**
**L. Lazzaroni**, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti
*IEEE Access* vol. 14, pp. 37416–37434 Rivista

**Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers**
A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 321–327 Conferenza

**Preface**
M. Ruo Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero
*Lecture Notes in Electrical Engineering* pp. v–vi Editoriale

**Optimizing Binary AutoEncoders on STM32 Microcontrollers**
H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 301–307 Conferenza

**Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students**
L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 242–248 Conferenza

**Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform**
H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 218–223 Conferenza

**Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA**
D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 25–31 Conferenza

**Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning**
O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 212–217 Conferenza

**Implementing YoloV8 on Zynq UltraScale+MPSoC FPGA Boards**
H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti
*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 289–294 Conferenza

**Generation of Synthetic Datasets of Highway Driving Scenarios**
M. Cossu, R. Berta, **L. Lazzaroni**, L. Forneris, M. Fresta, A. Pighetti, A. Pilla, J.L. Sauvaget, H. Touil, L. Durville, T. Griffon, G.B. Nejma, F. Hadj Selem, F. Bellotti
*IEEE Transactions on Intelligent Transportation Systems* Rivista

**FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving**
A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 295–301 Conferenza

**Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision**
A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 85–90 Conferenza

**Embedded Deployment of an LLM-Based Voice Assistant**
**L. Lazzaroni**, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 273–278 Conferenza

**Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection**
V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 286–291 Conferenza

**A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors**
S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 183–190 `Conferenza`

**A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems**
E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 249–254 `Conferenza`

**A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents**
A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 79–84 `Conferenza`

2025

**An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection**
**L. Lazzaroni**, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta
*IEEE Sensors Journal* vol. 25, no. 24, pp. 44859–44874 `Rivista`

**TinyML Acceleration with MAX78000**
A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta
*Proceedings of SIE 2024* Lecture Notes in Electrical Engineering pp. 468–474 `Conferenza`

**Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection**
L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 156–163 `Conferenza`

**Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking**
H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 387–394 `Conferenza`

**Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking**
F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 139–147 `Conferenza`

**RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset**
R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 395–402 `Conferenza`

**RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking**
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
*IEEE Sensors Letters* vol. 9, no. 9, art. 6010204 `Rivista`

**Quantization of MobileNetV2 for Resource-Constrained Microcontrollers**
**L. Lazzaroni**, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 497–504 `Conferenza`

**Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board**
H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 242–249 `Conferenza`

**On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection**
M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta
*IEEE Open Journal of the Industrial Electronics Society* vol. 6, pp. 1351–1369 `Rivista`

**Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices**
A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 323–330 `Conferenza`

**Green Machine Learning for the IoT (GEMINI): A Position Paper**
**L. Lazzaroni**, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 339–346 `Conferenza`

**Exploring Decision Transformer for Highway Automated Driving**
L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 123–130 `Conferenza`

**Deep Learning-Based Real-Time Driver Cognitive Distraction Detection**
M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta
*IEEE Access* vol. 13, pp. 26589–26607 `Rivista`

**Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection**
M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 250–257 `Conferenza`

**Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices**
A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti
*2025 IEEE International Conference on Consumer Electronics (ICCE)*
Digest of Technical Papers IEEE International Conference on Consumer Electronics `Conferenza`

**Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification**
A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 234–241 `Conferenza`

## 2024

**An embedded end-to-end voice assistant**
**L. Lazzaroni**, F. Bellotti, R. Berta
*Engineering Applications of Artificial Intelligence* vol. 136, art. 108998 `Rivista`

**Adversarial Policy Generation in Automated Parking**
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
*ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving* `Workshop`

**CBin-NN: An Inference Engine for Binarized Neural Networks**
F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti
*Electronics* vol. 13, no. 9, art. 1624 `Rivista`

**End-to-End Dataset Collection System for Sport Activities**
M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta
*Electronics* vol. 13, no. 7, art. 1286 `Rivista`

**YoloP-Based Pre-processing for Driving Scenario Detection**
M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 418–423 `Conferenza`

**Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection**
M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 444–449 `Conferenza`

**Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations**
A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 124–129 `Conferenza`

**Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity**
R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti
*Journal of Intelligent and Connected Vehicles* vol. 7, no. 3, pp. 229–244 `Rivista`

**Developing a TinyML Image Classifier in an Hour**
R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti
*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 946–960 `Rivista`

**Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving**
**L. Lazzaroni**, A. Pighetti, F. Bellotti, R. Berta
*Proceedings of SIE 2023* Lecture Notes in Electrical Engineering pp. 351–356 `Conferenza`

**Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge**
A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti
*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 781–794 `Rivista`

**Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach**
**L. Lazzaroni**, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 352–357 `Conferenza`

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**L. Lazzaroni**, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

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